NEERAJ VIJAYAKUMAR PATTANASHETTI

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EDUCATION

University of Illinois at Chicago (UIC) Expected: Dec 2026
Master of Science in Computer Science GPA: 4.0 / 4.0

SKILLS

Languages: Python | Java | SQL | C# | JavaScript
Agentic AI & ML: LangGraph | LLM Orchestration | MCP | RAG | NLP | PyTorch | TensorFlow
Backend & Data: FastAPI | Pydantic | REST APIs | OAuth | WebSockets | PostgreSQL | MySQL | Neo4j | Supabase
Cloud & Tools: AWS (EC2, S3, Lambda, API Gateway, IAM) | Docker | Git | Maven | Langfuse | Ollama

EXPERIENCE

SAP LABS, Palo Alto, USA **May, 2026 - Present**

AI Scientist Intern

- Built an **agentic evaluation system for enterprise AI agents**, integrating an agent evaluation harness with a structured rubric-generation pipeline spanning intent analysis, fact classification, and critic-based validation.
- Used evaluation results to drive **agent and skill optimization within the harness**, refining agent behavior and evaluation logic across retrieval, lookup, list, count, filtering, conditional, and time-sensitive tasks.
- Conducted deep **agent trace and failure analysis**, examining tool calls, retrieval behavior, generated requests, user-context propagation, and execution paths to distinguish agent failures from evaluation-system failures.
- Resolved critical **agent harness reliability issues** in per-request user context and metadata propagation, eliminating authentication and execution failures across enterprise evaluations.
- Designed evaluation strategies for **large structured JSON inputs**, using compact profiling, context control, and structured preprocessing to reliably evaluate agents with an **LLM judge**.

KreedaLoka, Bangalore, India **Jul, 2023 - Dec, 2023**

Software Engineering Intern

- Engineered **real-time backend systems** using Java, C#/.NET, and WebSockets, supporting 100+ concurrent users with low-latency state synchronization, gameplay updates, and reliable event-driven communication.
- Designed and optimized **data-intensive application logic**, including relational schemas, SQL queries, and dynamic tournament-matching workflows, reducing retrieval latency by 40% under concurrent load.

PROJECTS

Salesforce Metadata Impact Analysis System **Sep 2026**

- Built an agentic AI system **using LangGraph and LLM orchestration**, coordinating intent understanding, planning, deterministic tool execution, retrieval workflows, and response generation for multi-step metadata impact analysis.
- Designed a **compute-to-data architecture** with **Neo4j dependency graphs, structured tool interfaces, state/context management, and FastAPI**, integrating Pydantic validation, OAuth, Salesforce Metadata APIs, Supabase, and AWS for secure, explainable, production-grade agent execution.

Autonomous Agent Orchestration Platform **Jan 2026**

- Architected a supervision platform **for autonomous AI voice agents** using **locally hosted LLMs via Ollama** and a **Model Context Protocol (MCP) control plane** to manage agent discovery, call initiation, retrieval, orchestration, and lifecycle execution.
- Integrated **real-time telephony, transcription, RAG, and structured agent telemetry** to capture call outcomes, summaries, success signals, latency, and cost metrics, enabling human-in-the-loop analysis and iterative **agent evaluation**.